

Unit -5

Small Samples ($n < 30$)

Student's t-test:

→ Single mean: S.D is given directly

We know that

$$t = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n-1}}}$$

Degree of freedom = $n-1$

→ Single mean: SD is not given directly

We know that

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

$$\text{Where } s^2 = \frac{\sum (x - \bar{x})^2}{n-1} \quad (\text{or}) \quad \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2$$

→ Difference of mean:

$$t = \frac{\bar{x} - \bar{y}}{s \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \quad \frac{\bar{x} - \bar{y}}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}}$$

$$\text{Where } s^2 = \frac{1}{n_1 + n_2 + 2} \left[\sum (x - \bar{x})^2 + \sum (y - \bar{y})^2 \right] \quad (\text{or})$$

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 + 2}$$

(problem based on S.D given directly)

1. A Sample of 10 house owners is drawn and the following values of their incomes are obtained mean of Rs 6000. S.D Rs. 650 test the hypothesis that the average income of house owners of the town is Rs. 5500

Sol: Given $n=10$ ($n < 30$)

Sample mean $= \bar{x} = 6000$

S.D $= \sigma = 650$

$\mu = 5500$

1. $H_0: \bar{x} = \mu$

2. $H_1: \bar{x} \neq \mu$

3. $\alpha = 5\%$, Degree of freedom $= (n-1) = 10-1 = 9$

4. Test statistics

$$t = \frac{\bar{x} - \mu}{\frac{\sigma}{\sqrt{n-1}}} = \frac{6000 - 5500}{\frac{650}{\sqrt{10-1}}} = 2.31$$

5. table value of 5% of 9 is 1.833

6. Conclusion

calculated value > Table value

$$2.31 > 1.833$$

H_0 is rejected

problem based on (SD is not given directly)

1. A certain injection administered to each of 12 patients resulted in the following increases of blood pressure: 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. Can it be concluded that the injection will be accompanied by an increase in B.P?

Given $n = 12$

$$\text{mean} = \bar{x} = ?$$

$$S.D = S = \sigma = ?$$

population mean $\mu = ?$

We know that

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}}$$

x	5	2	8	-1	3	0	6	-2	1	5	0	4
x^2	25	4	16	1	9	0	36	4	1	25	0	16

$$\bar{x} = \frac{\sum x}{n} = \frac{31}{12} = 2.58$$

$$s^2 = \frac{\sum x^2}{n} - \left(\frac{\sum x}{n}\right)^2 = \frac{185}{12} - (2.58)^2 = 8.761$$

$$S = \sqrt{8.761} = 2.96$$

1. $H_0: \bar{x} = \mu$

2. $H_1: \bar{x} \neq \mu$

3. $\alpha = 5\%$. D.O.F = $n - 1 = 12 - 1 = 11$

4. Test statistics

$$t = \frac{\bar{x} - \mu}{\frac{S}{\sqrt{n}}} = \frac{2.58 - 0}{\frac{2.96}{\sqrt{11}}} = 2.89$$

5. Table value $\alpha = 5\% = 1.796$

6. Conclusion:

(1) Calculated value > Table value

$$2.89 > 1.796$$

H_0 is rejected

problem based on Fitterness of mean

1. We know that

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{S^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}}$$

$$S^2 = \frac{\sum (x - \bar{x}_1)^2 + \sum (x_2 - \bar{x}_2)^2}{n_1 + n_2 - 2} \quad (\text{or})$$

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$S^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2}$$

$$\text{Where } s_1^2 = \frac{\sum x_1^2}{n_1} - \left(\frac{\sum x_1}{n_1} \right)^2$$

$$s_2^2 = \frac{\sum x_2^2}{n_2} - \left(\frac{\sum x_2}{n_2} \right)^2$$

1. Two horses A and B were tested according to the time (in seconds) to run a particular race with the following results.

Horse A: 28 30 32 33 33 29 34

Horse B: 29 30 30 24 27 29

Test whether the horse A is running whether the B at 5% loss.

Sol $n_1 = 7$ $n_2 = 6$

$\bar{x}_1 = ?$ $\bar{x}_2 = ?$

$$x_1: 28 \quad 30 \quad 32 \quad 33 \quad 33 \quad 29 \quad 34 \quad \Sigma x_1 = 219$$

$$\Sigma x_1^2 = 6883$$

$$x_2: 29 \quad 30 \quad 30 \quad 32 \quad 22 \quad 29 \quad \Sigma x_2 = 169$$

$$\Sigma x_2^2 = 4789$$

$$\bar{x}_1 = \frac{\Sigma x_1}{n} = \frac{219}{7} = 31.29$$

$$\bar{x}_2 = \frac{\Sigma x_2}{n} = \frac{169}{6} = 28.17$$

$$s_1^2 = \frac{\Sigma x_1^2}{n} - \left(\frac{\Sigma x_1}{n}\right)^2 = \frac{6883}{7} - (31.29)^2 = 4.23$$

$$s_2^2 = \frac{\Sigma x_2^2}{n} - \left(\frac{\Sigma x_2}{n}\right)^2 = \frac{4789}{6} - (28.17)^2 = 4.28$$

$$s^2 = \frac{n_1 s_1^2 + n_2 s_2^2}{n_1 + n_2 - 2} = \frac{7(4.23) + 6(4.28)}{7 + 6 - 2} = 5.03$$

$$1. H_0 = \bar{x}_1 = \bar{x}_2$$

$$2. H_1 = \bar{x}_1 \neq \bar{x}_2$$

$$3. \alpha = 5\%. \quad \text{DOF} = n_1 + n_2 - 2 = 7 + 6 - 2 = 11$$

4. Test statistics

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s^2 \left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} = \frac{31.29 - 28.17}{\sqrt{5.03 \left(\frac{1}{7} + \frac{1}{6}\right)}} = 2.498$$

$$5. \text{Table value of } 5\% \text{ in } 11 = 1.796$$

6. Conclusion

$$\text{Cal value} > \text{Table value}$$

$$2.498 > 1.796$$

$\therefore H_0$ is rejected.

F-Test

We know that

$$F = \frac{\text{Greater Variance}}{\text{Small Variance}} = \frac{S_1^2}{S_2^2} \text{ or } \frac{S_2^2}{S_1^2}$$

$$\text{Where } S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}, S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1}$$

Here S_1^2 & S_2^2 are variance and S_1 & S_2 are S.D.

The degree of Freedom ($n_1 - 1, n_2 - 1$)

Note:

→ t-test depends on mean

→ F-test depends on variance

1. A group of 10 rats feed on diet A and another group of 8 rats feed on diet B recorded the following increase in weight

Diet A: 5 6 8 1 12 4 3 9 6 10

Diet B: 2 3 6 8 10 1 2 8

Find if the variances are significant different

Sol Given $n_1 = 10 (< 30)$ $n_2 = 8 (< 30)$

We know that

$$F\text{-test} = \frac{S_1^2}{S_2^2} \text{ or } \frac{S_2^2}{S_1^2} = \frac{\text{Greater value}}{\text{Smallest value}}$$

We know that

$$\sigma_1^2 = S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1}, \quad \sigma_2^2 = S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1}$$

x	x - $\bar{x}$	y	(y - $\bar{y}$)	(x - $\bar{x}$) ²	(y - $\bar{y}$) ²
5	-1.4	2	-3	1.96	9
6	-0.4	3	-2	0.16	4
8	2.6	6	-1	2.56	1
1	5.4	8	3	29.16	9
12	5.6	10	5	31.36	25
4	-2.4	1	-4	5.76	16
3	-3.4	2	-3	11.56	9
9	2.6	8	3	6.76	9
6	-0.4			0.16	
10	3.6			12.96	
<u>64</u>		<u>40</u>		<u>102.4</u>	<u>82</u>

$$\bar{x} = \frac{\sum x}{n} = \frac{64}{10} = 6.4, \quad \bar{y} = \frac{\sum y}{n} = \frac{40}{8} = 5$$

$$S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1} = \frac{102.4}{10 - 1} = \frac{102.4}{9} = 11.37$$

$$S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1} = \frac{82}{8 - 1} = \frac{82}{7} = 11.714$$

$$F = \frac{S_1^2}{S_2^2} = \frac{11.37}{11.714} \text{ or } \frac{S_2^2}{S_1^2} = \frac{\text{Greater value}}{\text{Small value}}$$

$$= \frac{11.714}{11.37} = 1.0302$$

$$\text{degree of freedom} = (n_1 - 1, n_2 - 1)$$

$$= (10 - 1, 8 - 1)$$

$$= (9, 7) = 3.68 \text{ Table value}$$

$$1. H_0 = \sigma_1^2 = \sigma_2^2$$

$$2. H_1 = \sigma_1^2 \neq \sigma_2^2$$

$$3. \alpha = 0.05 \text{ 5\%}$$

$$\text{Degree of freedom } (n_1 - 1, n_2 - 1) = (9, 7)$$

$$= 3.68 \text{ Table Value}$$

4. Test statistics

$$F = 1.0302$$

5. Conclusion

Calculated value < Table value

$$1.0302 < 3.68$$

$\therefore H_0$ is accepted

2. In one sample of 10 observations the sum of the squares of the deviations of the sample values from the sample mean was 120 and in another sample of 12 observation it was 314. Test when their the difference is significant at 5% loss

$$n_1 = 10 (< 30) \quad n_2 = 12 (< 30)$$

$$\sum (x - \bar{x})^2 = 120, \quad \sum (y - \bar{y})^2 = 314$$

$$S_1^2 = \frac{\sum (x - \bar{x})^2}{n_1 - 1} = \frac{120}{10 - 1} = \frac{120}{9} = 13.33$$

$$S_2^2 = \frac{\sum (y - \bar{y})^2}{n_2 - 1} = \frac{314}{12 - 1} = \frac{314}{11} = 28.54$$

$$F = \frac{S_1^2}{S_2^2} \text{ or } \frac{S_2^2}{S_1^2} = \frac{28.54}{13.33} = 2.141$$

$$\text{degree of freedom} = (n_1 - 1, n_2 - 1)$$

$$= (10 - 1, 12 - 1)$$

$$= (9, 11)$$

$$= 2.90 \text{ (Table value)}$$

$$1. H_0 = \sigma_1^2 = \sigma_2^2$$

$$2. H_1 = \sigma_1^2 \neq \sigma_2^2$$

$$3. \alpha = 5\%$$

$$(9, 11)$$

$$\text{degree of freedom} = 2.90 \text{ Table value}$$

4. Test statistics

$$F = 2.141$$

5. Conclusion

calculated value < Table value

$$2.141 < 2.90$$

H_0 is accepted

* Chi Square Test

1. Goodness of fit

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Where O - Observed Frequency

E = Expected Frequency

$$\text{Degree of freedom} = (n - 1)$$

2. Independence of attributes

a	b	a+b
c	d	c+d
a+c	b+d	$N = a+b+c+d$

degree of freedom
 $= (r-1)(s-1)$
 r - rows number
 s - no of columns

The expected Frequencies are given by

$\frac{(a+b) \times (a+c)}{N}$	$\frac{(a+b) \times (b+d)}{N}$	a+b
$\frac{(c+d) \times (a+c)}{N}$	$\frac{(c+d) \times (b+d)}{N}$	c+d
a+c	b+c	a+b+c+d = N

1. The following table gives the number of air craft accidents that occurred. Various various days of the week during whether the accidents are uniformly distributed over the week

Days	Mon	Tue	Wed	Thue	Fri	Sat
No of accidents	14	18	12	11	15	14

Sol: H_0 : The accidents are uniformly distributed over the week

2. H_1 : The accident are not uniformly distributed over the week

3. $\alpha = 5\%$

Degree of freedom = $n - 1$

= $6 - 1 = 5 = 11.07$ Table value

4. Test statistics

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Expected value = $E = \frac{\sum O}{n} = \frac{84}{6} = 14$

O	E	(O - E)	(O - E) ²	$\frac{(O - E)^2}{E}$
14	14	0	0	0
18	14	4	16	1.143
12	14	-2	4	0.286
11	14	-3	9	0.689
15	14	1	1	0.091
14	14	0	0	0
				$\chi^2 = 2.148$

$\chi^2 = 2.148$

5. Conclusion

Calculated value < Table value

$2.148 < 11.07$

6. H_0 is accepted

2. The theory predicts the proportion of bears in four groups A, B, C, & D should be 9:3:3:4. In an experiment with 1600 bears the numbers in the four groups were 882, 313, 287 & 118. Does the experiment result support the theory (Given

<u>Sol</u>	O	E	O-E	$(O-E)^2$	$\frac{(O-E)^2}{E}$
882	900	-18	324	0.360	
313	300	13	169	0.563	
287	300	-13	169	0.563	
118	100	18	324	3.240	
				<u>4.726</u>	

$$9:3:3:1 = 1600$$

$$\frac{9}{16} \times 1600 = 900$$

$$\frac{3}{16} \times 1600 = 300$$

$$\frac{1}{16} \times 1600 = 100$$

1. H_0 : The experiment support the theory
2. H_1 : The experiment result not support the theory
3. $\alpha = 5\%$

$$\text{degree of freedom} = n-1 = 4-1 = 3 = 7.81$$

4. Test statistics (table value)

$$\chi^2 = 4.726$$

5. Conclusion

$$\text{Calc. value} < \text{Table value}$$

$$4.726 < 7.81$$

H_0 is accepted

Intermediate

Independence of attributes

1. Given the following Contingency table hair color and eye color. Find the value of χ^2 . Is there good association between the two

Eye color	Hair color			Total
	Blue	Brown	Black	
Blue	15	5	20	40
Grey	20	10	20	50
Brown	25	15	20	60
Total	60	30	60	150

Sol Expected Frequency are given below

15	5	20	40
20	10	20	50
25	15	20	60
60	30	60	150

$\frac{40 \times 60}{150} = 16$	$\frac{40 \times 30}{150} = 8$	$\frac{40 \times 60}{150} = 16$	40
$\frac{50 \times 60}{150} = 20$	$\frac{50 \times 30}{150} = 10$	$\frac{50 \times 60}{150} = 20$	50
$\frac{60 \times 60}{150} = 24$	$\frac{60 \times 30}{150} = 12$	$\frac{60 \times 60}{150} = 24$	60
60	30	60	150

O	E	O-E	(O-E) ²	$\frac{(O-E)^2}{E}$
15	16	-1	1	1/16
5	8	-3	9	9/8
20	16	4	16	16/16
20	20	0	0	0/20
10	10	0	0	0/10
20	20	0	0	0/20
25	24	1	1	1/24
15	12	3	9	9/12
20	24	-4	16	16/24
				3.6458

$$\chi^2 = 3.6458$$

$$\text{Degree of freedom} = (r-1)(s-1)$$

$$= (3-1)(3-1)$$

$$= (2)(2)$$

$$= 4 = 9.488 \text{ } 5\%$$

Conclusion

Cal value < Table value

$$3.6458 < 9.488$$

H₀ is accepted

↓ Table value

Paired T-test

5%, 95% confidence interval

Before	86	92	100	93	88	80	88	72	95	106
After	77	84	92	87	80	74	80	85	95	96

1. $H_0 = \mu_1 = 0$ no significant difference between the after and before reduction programme

2. $H_1 = \mu_1 \neq 0$ is a significant difference between the after and before weight reduction program

3. $\alpha = 5\% \Rightarrow 0.05$

Calculation

x_1 denotes the before weight reduction program
 x_2 denotes the after weight reduction programme

$$t = \frac{|\bar{d}|}{s/\sqrt{n}} \quad \bar{d} = \frac{\sum d}{n}, \quad s = \sqrt{\frac{\sum (d - \bar{d})^2}{n-1}}$$

x_1	x_2	$d = x_1 - x_2$	$d - \bar{d} (\bar{d} = 70)$	$(d - \bar{d})^2$
86	77	9	2	4
92	84	8	1	1
100	92	8	1	1
93	87	6	-1	1
98	80	8	1	1
80	74	6	-1	1
88	80	8	1	1
92	85	7	0	0
95	95	0	-7	49
106	96	10	3	9
		70		68

$$\bar{d} = \frac{\sum d}{n} = \frac{70}{10} = 7$$

$$S = \sqrt{\frac{\sum (d - \bar{d})^2}{n-1}} = \sqrt{\frac{68}{10-1}} = \sqrt{\frac{68}{9}} = 2.748$$

$$t = \frac{|\bar{d}|}{\frac{S}{\sqrt{n}}} = \frac{7}{\frac{2.75}{\sqrt{10}}} = \frac{7}{0.87} = \underline{\underline{8.05}}$$

Level of Significance

$$\alpha = 5\% \text{ one tailed test} = 5\%$$

$$= 10\%$$

$$= 0.1$$

$$\text{Degree of freedom} = n-1$$

$$= 10-1 = 9 \text{ at } 10\% \text{ or } 0.1$$

↓
1.383 Table value

Conclusion

Calculated value > Table value

$$8.05 > 1.383$$

H₀ is reject

95% and 99% Confidence intervals

$$95\% = t = \frac{|\bar{d}|}{\frac{S}{\sqrt{n}}}$$

$$|\bar{d}| \pm t\text{-table value} \times \frac{S}{\sqrt{n}} \rightarrow 0$$

$$\text{Given} = 95\% \Rightarrow 100 - 95\% = 5\%$$

Level of Significance $\alpha = 5\% = 0.05$

degree of freedom $= n-1 = 10-1 = 9$ Table Value $= 1.833$

From ① $\Rightarrow (\bar{d}) \pm t \text{ table value} \times \frac{s}{\sqrt{n}}$

$$7 \pm 1.833 \times \frac{2.748}{\sqrt{10}}$$

$$7 \pm 1.833 \left(\frac{2.748}{0.87\sqrt{10}} \right)$$

$$7 \pm 1.833 (0.87)$$

$$7 + 1.833(0.87), 7 - 1.833(0.87)$$

$$7 + 1.595 (0.87) 7 - 1.595$$

$$8.595 (0.87) 5.405$$

The population values lies b/w the 8.595 and 5.405.

$$99\% = t = \frac{(\bar{d})}{\frac{s}{\sqrt{n}}}$$

$$(\bar{d}) \pm t \text{ table value} \times \frac{s}{\sqrt{n}} \rightarrow \textcircled{2}$$

$$\text{Given } 99\% \Rightarrow 100 - 1\% = 99\%$$

$$\underline{1\% = 0.01}$$

Level of Significance $= 1\%$

degree of freedom $= n-1 = 10-1 = 9$
 $= 2.821$ Table value

From ② $(\bar{d}) \pm t \text{ table value} \times \frac{s}{\sqrt{n}}$

$$7 \pm 2.821 \times (0.87)$$

$$7 \pm 2.45427$$

$$7 + 2.46, 7 - 2.46$$

$$9.46(9) 4.54$$

The population values lie between 9.46 and 4.54